

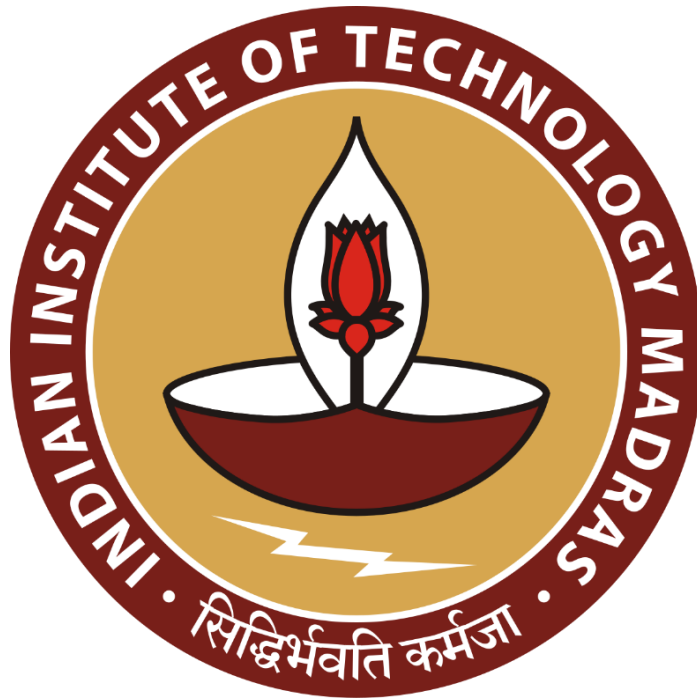
DATA-DRIVEN ANALYSIS OF POULTRY FARM OPERATIONS AND LIVE CHICKEN SUPPLY

A Mid-Term report for the BDM capstone Project

Submitted by

Name: PULLURI YASHWANTH KRISHNA

Roll number: 23f1002174



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

Contents

1 Executive Summary	2
2 Proof of originality of the Data	2
3 Metadata	5
Information about the Chick Mortality Data	5
Information about the Sales Data	7
Information about the Inventory Data	9
Descriptive Statistics	9
4 Detailed Explanation of Analysis Process/Method	10
5 Results and Findings	11

1 Executive Summary

This project is based on Karthik Reddy Poultry, a small-scale B2B poultry farm located in Chinna Masanpally village of Rayapole Mandal, Siddipet District, Telangana. The farm was started in 2023 by Mr. Karthik Reddy and supplies live chickens to nearby meat shops, hotels, and restaurants. The farm operates regularly with multiple batches and a small workforce, it faces several operational problems that affects its overall performance. One major issue is high chick mortality, mainly due to poor temperature regulation, irregular vaccination practices, and improper hygiene practices. The farm experiences inconsistent sales, which affects revenue stability. Poor inventory management on feed, medicines, and vaccines further adds financial stress because of wastage and unnecessary expenses.

To study these problems, primary data was collected from the farm's day-to-day operations through manual records, bills, and discussions with the owner. Daily records related to chick mortality and sales, and monthly information on inventory usage is collected for July to September. Key variables such as shed temperature, humidity, vaccination status, number of birds sold, selling price, buyer type, stock consumption, expiry details were considered. Basic Descriptive statistics are used to summarize data and identify trends and variations related to mortality, sales performance, and inventory usage.

To address these challenges, regression analysis is used to understand the factors influencing chick mortality, time-series analysis is applied to study sales patterns over time, and ABC analysis is used to classify inventory items based on consumption value. Improved hygiene and vaccination led to lower mortality levels, sales show gradual improvement over 3-month period, and reduced inventory wastage through better prioritization. These mid term findings highlight the potential of data-driven decision-making in improving operational efficiency at Karthik Reddy Poultry.

2 Proof of originality of the Data

- Business Name: Karthik Reddy Poultry
- Address: Chinna Masanpally Village, Rayapole Mandal, Siddipet District, Telangana (502278)
- Owner's Name: Mr. Karthik Reddy

Video of interaction with Business owner and Letter from the Poultry Farm:

<https://drive.google.com/drive/folders/1ZsQzRXFA9x4W8ZZUfXPyhWjINgi0StTT?usp=sharing>



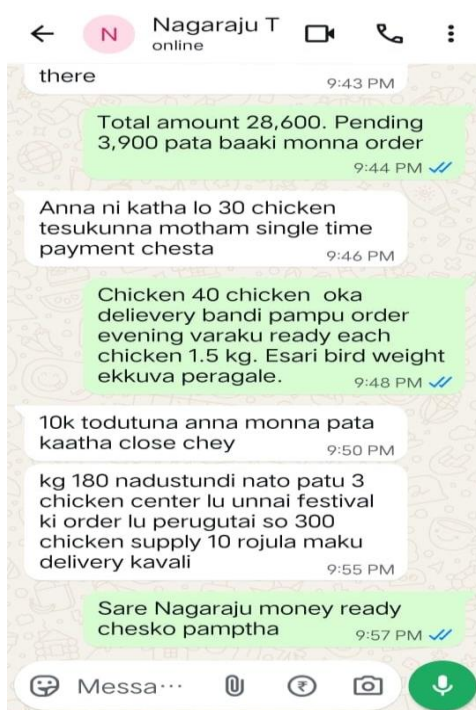
Fig 1. Me along with the Business Owner at the Business place.

Images Folder Link:

<https://drive.google.com/drive/folders/1Y7kN0AGFizfY1xnaB3mZPNBgsK7tHsvV?usp=sharing>

The data used for this project is primary in nature and was collected from business owner of Karthik Reddy Poultry. The Sources I used for Data Collection are:

1. **Verbal Interaction with Business Owner:** Most of operational information related to chick mortality, vaccination practices, sales quantities and inventory usage was collected through communication with the business owner.
2. **Daily Notepad:** Information such as chick deaths, vaccination activities, Sales details and inventory ordered and used are noted in farm Daily Notepad.
3. **WhatsApp Messages:** Used for buyers regarding orders, confirmations and payment-related information. These messages are mostly evidence of transaction-related details. Sales quantities are noted through WhatsApp messages.



AUGUST BATCH		10-8-2025
1 Bag of New Life	₹ 25,500	
2 Bags of Chicken	₹ 43,000	
Glucose	₹ 400	
Adrenal	₹ 400	
4 P/Disinfectant	₹ 15,000	
8-shock Vaccine (ND)	₹ 12,000	
4-shaker feed	₹ 45,000	
shed	₹ 12,000	
Nestling	₹ 15,000	
Bird feed	₹ 75,000	
Kagaru	₹ 15,000	(Paid 15,000)
Amranyale	₹ 5,000	(Paid 5,000)
Amranyale	₹ 45,51	(Paid 45,51)
Amranyale	₹ 45,75	(Paid 45,75)
Burdu	₹ 9,000	
Yellal	₹ 6,250	
Kamath	₹ 12,000	

Fig 2. Snapshots of data recordings

3 Metadata

3.1 Information about the Chick Mortality Data

The chick mortality dataset contains day-wise recorded collected from manual poultry farm registers at Karthik Reddy Poultry Farm. The data was recorded to monitor chick health, environmental conditions, and operational practices

Dataset Link: [Chick Mortality Data](#)

Data Collection Duration: 3 Months

- Range: 1 July 2025 to 30 September 2025
- Data Format: Excel Sheets (XLSX)
- Units of Measurement:
 - Temperature: Degree Celsius
 - Humidity: Percentage (%)
 - Quantity: Number of birds / items
 - Weight: Kilograms (kg)
 - Money: Indian Rupees (₹)

Dataset Size:

- Number of rows: 92 days records

- Total No. of Columns: 24
- Relevant columns used for analysis: 11

Variable Name	Description	Data Type	Relevance to Problem
Date	Date of observation	Date	Enables time-based mortality analysis
Batch_Id	Batch identifier	Categorical	Helps track batch-wise mortality
Chick_Age_Days	Age of chicks in days	Numeric	Mortality varies with age
Opening_Chick_Count	Number of chicks at start of day	Numeric	Base population for mortality rate
Chicks_Died_Today	Daily chick deaths	Numeric	Direct indicator of loss
Cumulative_Deaths	Total deaths till date	Numeric	Shows severity over time
Mortality_Rate	Percentage mortality per day	Numeric	Key business risk metric
Shed_Temperature_Celsius	Temperature inside shed	Numeric	Affects chick health
Humidity	Shed humidity level	Numeric	Influences disease spread
Ventilation_Status	Quality of ventilation	Categorical	Operational hygiene indicator
Litter_Condition	Floor condition	Categorical	Hygiene-related mortality factor

Fig 3. Description of Key features in Chick Mortality Dataset

3.2 Information about the Sales Data

The sales data was recorded from Karthik Reddy poultry through daily notepad and WhatsApp messages from the business owner. The sales data records day-wise live-bird sales and revenue information of the poultry farm. The data helps track demand patterns and sales performance over time.

Dataset Link: [Sales Data](#)

Data Collection Duration: 3 Months

- Range: 1 July 2025 to 30 September 2025
- Data Format: Excel Sheets (XLSX)

Dataset Size:

- Number of rows: 92 days records
- Total No. of Columns: 16
- Relevant columns used for analysis: 13

Variable Name	Data Type	Description	Relevance to Problem
Date	Date	Sales transaction date	Time-series sales analysis
Month	Categorical	Month of sale	Monthly demand comparison
Week_Number	Numerical	Week identifier	Weekly sales trends
Invoice_Number	Categorical	Invoice ID	Transaction tracking
Birds_Available	Numerical	Birds available for sale	Supply capacity
Birds_Sold	Numerical	Birds sold per day	Demand measurement
Unsold_Birds	Numerical	Unsold inventory	Demand–supply mismatch
Average_Bird_Weight_kg	Numerical	Average weight per bird	Pricing influence
Selling_Price_per_kg	Numerical (₹)	Price per kg	Revenue driver
Total_Sales_Amount	Numerical (₹)	Daily revenue	Business performance metric
Buyer_Type	Categorical	Type of buyer	Market segmentation
Buyer_Location	Categorical	Buyer location	Demand geography
Repeat_Customer	Categorical	Repeat purchase indicator	Customer retention

Fig 4. Description of Key features in Sales Dataset

3.3 Information about the Inventory Data

The inventory dataset captures usage and cost details on feed, medicines, vaccines and used in farm operations. The inventory data is collected from farm notepads and interactions with business owner.

Dataset Link: [Inventory Data](#)

Data Collection Duration: 3 Months

- Range: 1 July 2025 to 30 September 2025
- Data Format: Excel Sheets (XLSX)

Dataset Size:

- Number of rows: 19
- Total No. of Columns: 16
- Relevant columns used for analysis: 8

Variable Name	Description	Data Type	Relevance to Problem
Item_Id	Inventory item ID	Categorical	Item identification
Item_Name	Name of item	Categorical	Cost tracking
Category	Item category	Categorical	Inventory classification
Supplier_Name	Supplier details	Categorical	Procurement analysis
Opening_Stock_Quantity	Stock at start	Numeric	Inventory control
Purchased_Quantity	Quantity purchased	Numeric	Replenishment
Consumed_Quantity	Quantity consumed	Numeric	Usage analysis
Closing_Stock_Quantity	Remaining stock	Numeric	Waste detection
Unit_Cost_INR	Cost per unit	Numeric	Cost calculation
Total_Consumption_Value	Total cost consumed	Numeric	Expense impact
Expiry_Date	Expiry date	Date	Wastage risk

Fig 5. Description of Key features in Inventory Dataset

3.4 Descriptive Statistics

Statistic	Total_Consumption_Value
Count	18
Mean	7880.277778
Median	6637.5
Minimum	1350
Maximum	18240
25%	3075
50%	6637.5
75%	11400
Standard deviation	5548.763103

Fig 6. Descriptive statistics of Total Inventory Consumption Value

Statistic	Consumed_Quantity	Total_Consumption_Value
Count	18	18
Mean	175.8333333	7880.277778
Median	90	6637.5
Minimum	30	1350
Maximum	480	18240
25%	61.25	3075
50%	90	6637.5
75%	282.5	11400
Standard deviation	150.7164264	5548.763103

Fig 7. Descriptive statistics of Inventory Consumption Quantity and Value

Statistic	Birds_Sold	Total_Sales_Amount
Count	92	92
Mean	32.70652174	9298.804348
Median	32	8384
Minimum	17	2992
Maximum	48	17734
25%	24	5046.75
50%	32	8384
75%	44	14036
Standard deviation	9.685763931	4618.693132

Fig 8. Descriptive statistics of Sales Volume and Total Sales Amount

	count	mean	std	min	25%	50%	75%	max
Item_Name								
Broiler Feed	3.0	446.666667	35.118846	410.0	430.0	450.0	465.0	480.0
Disinfectant	3.0	45.000000	13.228757	30.0	40.0	50.0	52.5	55.0
Litter Material	3.0	140.000000	45.825757	90.0	120.0	150.0	165.0	180.0
Starter Feed	3.0	283.333333	20.816660	260.0	275.0	290.0	295.0	300.0
Vaccines(ND)	3.0	81.666667	10.408330	70.0	77.5	85.0	87.5	90.0
Vitamins	3.0	58.333333	7.637626	50.0	55.0	60.0	62.5	65.0

Fig 9. Item-wise Descriptive Statistics of Inventory Consumption Value

	count	mean	std	min	25%	50%	75%	max
Month								
August	30.0	8414.97	1069.40	6300.0	7649.5	8384.0	8910.0	11172.0
July	31.0	4347.10	789.44	2992.0	3683.5	4322.0	4993.5	6084.0
September	30.0	15018.23	1356.15	11760.0	14036.0	15266.5	15870.0	17280.0

Fig 10. Month-wise Descriptive Statistics of Total Sales Amount

Statistic	Chicks_Died_Today	Mortality_Rate
Count	92	92
Mean	1.656	0.626
Median	1.35	0.44
Minimum	0	0
Maximum	4	1.82
25%	1	0.34
50%	2	0.67
75%	2	0.885
Standard deviation	1.133	0.455

Fig 11. Descriptive Statistics of chick mortality data

Linkage of Data Collected to Problem Statements

I used Python(Pandas and Numpy) for computing descriptive statistics such as mean, median standard deviation, minimum, maximum, and quartiles for the selected variables across chick mortality, sales, and inventory datasets.

- **Chick mortality records:** The mortality data, including daily chick deaths, mortality rate, temperature, humidity, and hygiene conditions. addresses the first problem by identifying operational and environmental factors contributing to chick losses.
- **Sales records:** The sales data, including birds sold, selling price, and total sales amount, supports the second problem by offering a detailed view of demand fluctuations and revenue variability. Sales records enable analyze trend patterns and help in optimal production planning.
- **Inventory Cost Management:** The inventory data, including consumed quantity and total consumption value, addresses the third problem statement by highlighting high-cost and high-usage items. This forms the basis for inventory optimization and cost control using ABC analysis.

4 Detailed Explanation of Analysis Process & Methods

4.1 Data Collection

The data used for analysis was collected from daily operational data and interaction with business

owner. Farm maintained most of its sales data in WhatsApp messages.

The datasets used are:

- Chick mortality data(daily operational and environmental records)
- Sales data (daily sales invoices and transaction logs)
- Inventory data (purchase and consumption records of feed, medicines, and vaccines.

The data was collected on a daily basis for a period of three months (July 2025 to September 2025) and stored in CSV and Excel formats. These datasets form the basis for analyzing mortality loss, revenue instability, and inventory cost inefficiencies.

4.2 Data Cleaning and Pre-Processing

The collected data contained inconsistencies such as missing values, data entry errors and inconsistent categorical labels. Data cleaning ensured accuracy, consistency, and reliability of analysis.

The following preprocessing steps were carried out:

- Missing values were identified and handled appropriately.
- Duplicate and inconsistent records were removed.
- Categorical variables such as Ventilation_Status, Litter_Condition, Buyer_Type, and Repeat_Customer were standardized.
- Numeric variables were checked for outliers and corrected where necessary.
- Derived variables such as Mortality_Rate, Unsold_Birds, and Total_Consumption_Value were verified for correctness.

Python libraries such as Pandas and NumPy were used for data cleaning and preprocessing.

4.3 Analysis Process and Methods

Regression Analysis(Chick mortality): Regression analysis is used to study the impact of environmental and operational factors such as shed temperature, humidity, ventilation status, and litter condition on chick mortality. This method is suitable as mortality is influenced by multiple factors and requires quantitative relationship analysis.

Dependent Variable: Mortality_Rate

Independent Variables: Shed_Temperature_Celsius, Humidity, Ventilation_Status, Litter_conditon etc.

Time-Series Analysis (Sales Data): Time-series analysis is used to analyze daily sales and revenue trends over time. This method helps identify demand fluctuations and sales instability, supporting better batch planning and pricing decisions.

ABC Analysis (Inventory Data):

ABC analysis is used to classify inventory items based on their total consumption value into high, medium, and low priority categories. This method supports inventory cost control and wastage reduction

5 Results and Findings

5.1 Chick Mortality Results

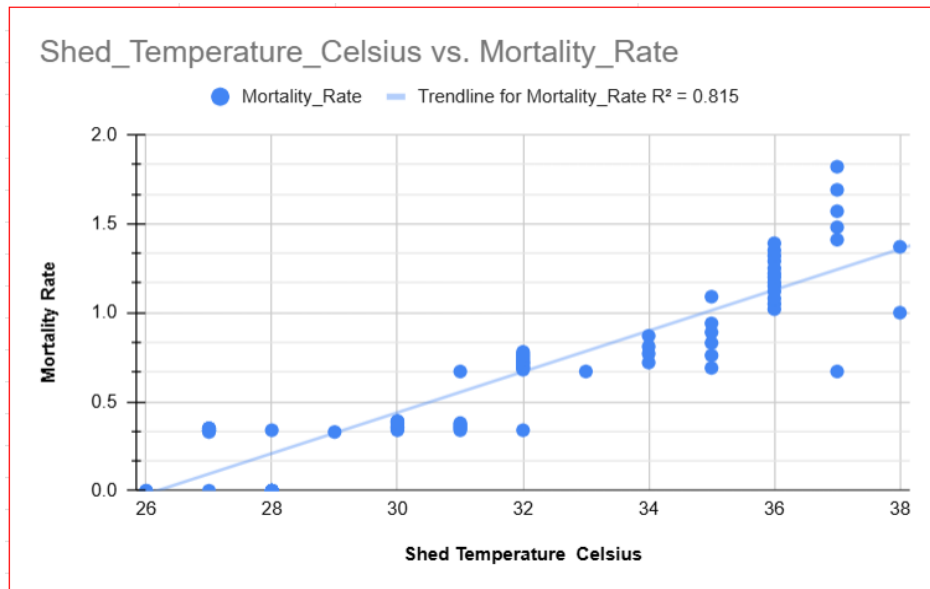


Fig 12. Relationship between Shed Temperature and Chick Mortality Rate (Regression Analysis)

- Regression analysis shows a strong positive relationship between shed temperature and chick mortality rate.
- As shed temperature increases, chick mortality also increases.
- The R2 value of 0.815 indicate that shed temperature explains about 81.5% of the variation in mortality.
- Proper temperature regulation and shed management can significantly reduce mortality and can reduce revenue losses in Karthik Reddy poultry.

5.2 Sales Results

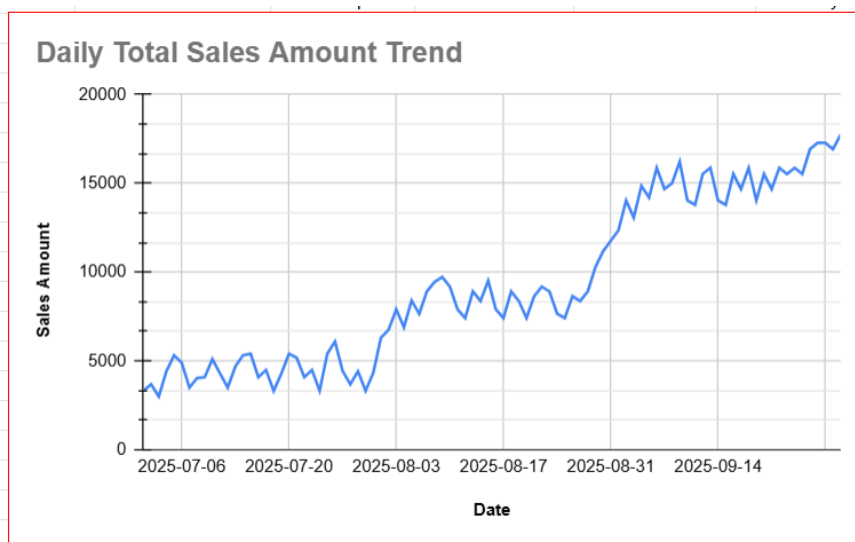


Fig 13. Daily Total Sales Amount (Time-Series Analysis)

- The daily sales trend shows a clear upward movement from July to September, indicating increasing demand.
- Sales fluctuations are observed on a daily basis, reflecting variation in customer arrivals and batch sales.
- The increasing trend assures the effectiveness of better batch planning and improvement in shed management.

5.3 Inventory Results

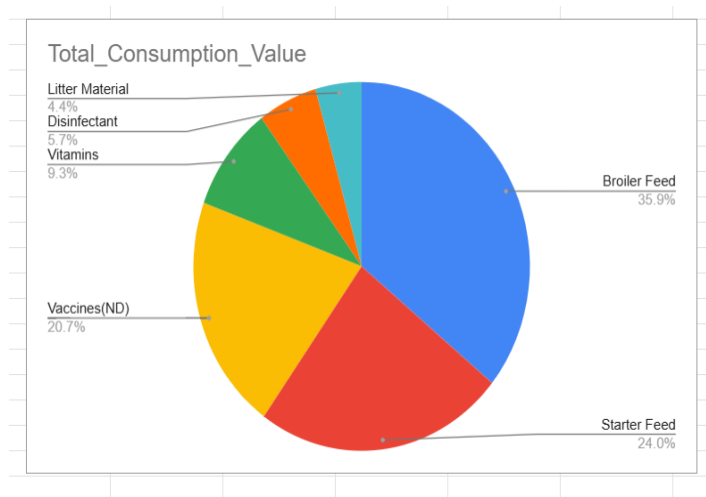


Fig 13. Inventory Consumption Value distribution by Item (ABC Analysis)

- ABC analysis based on total consumption value shows that Broiler Feed and Starter Feed are Class A items, contributing the highest inventory cost.
- Vaccines (ND) fall under Class B, while Vitamins, Disinfectant, and Litter Material are Class C items with lower cost impact. This results supports priority control of feed inventory to reduce wastage and inventory cost efficiency.